

Web Application Design Specs for the Ukraine Population Project

The Demographic Science Unit in the Department of Population Health at the University of Oxford is requesting quotations for developing an interactive web mapping tool to display real-time population estimates produced by the Ukraine Population Project. The design specs are listed below to facilitate an informed discussion prior to generating a quotation for the work.

Deadlines

Functional prototype: 1 March 2024

Final product: 25 June, 2024

Front-end

- Full screen interactive Leaflet map showing Ukraine population estimates for sub-national administrative units, including Crimea and all occupied territories (see figure below)
- Control panel to select user options and Results panels to display graphed results that compliment the map (see figure below)

Controls:

- o Slider to select the date of population estimates (Feb 24, 2022 until current date)
- o Drop-down to select sex (all, male, female)
- o Range slider to select age groups (i.e. 5-year age group options from 0 to 80+)
- o Drop-down to select geographic level (Oblasts, Raions, Hromadas. Etc.)

Map:

- o Administrative units coloured on the map to indicate population size
- o Hovering over administrative unit shows its name and population estimate
- o Clicking the map updates all results to represent the selected geographic unit

Results:

- o Population panel: Probability distribution of population estimate. X-axis = Population; Y-axis = Probability
- o Age-sex panel: Population pyramid plot to show age-sex proportions
- o Migration panel: Chord chart to show migration probabilities

Potential front-end features

- o Ability to toggle on/off other map layers (e.g. conflict event locations, road networks, border crossings)

Back-end

- **Dockerised PostgreSQL database** to store population estimates, migration probabilities, and other dashboard data
- **Dockerised Flask API** to allow the front-end to query the database
- **Dockerised Nginx** web server

Please follow-up to schedule a meeting, if needed, to discuss the project and answer any questions that you may have before generating a quotation.

Please prepare cost estimates for the front-end and back-end development work separately in your quotation. Send quotation(s) to douglas.leasure@demography.ox.ac.uk.

